ensuring that drones become safer, smarter, and more integrated into human society.

7.11.1 Emerging technologies addressing energy, navigation, and autonomy issues

Energy efficiency, precise navigation, and autonomous decision-making remain three of the most critical areas of improvement in drone technology. New advances are paving the way for longer endurance, greater precision, and higher operational intelligence.

In the field of energy and propulsion, innovations such as hydrogen fuel cells, solar-powered systems, and hybrid-electric propulsion are extending flight duration while reducing carbon emissions. Solid-state batteries and graphene-based supercapacitors promise higher energy density and faster charging, addressing one of the biggest limitations of current UAVs: short flight times. Additionally, wireless charging pads and mid-air power transmission systems are under research to enable continuous operations without frequent battery replacements.

In navigation, progress is being made to overcome the limitations of GPS dependency. AI-driven visual odometry, LiDAR-based mapping, and simultaneous localization and mapping (SLAM) technologies now allow drones to operate accurately in GPS-denied environments, such as indoors or under dense canopies. The integration of quantum sensors and inertial navigation systems (INS) further enhances spatial awareness and precision.

Regarding autonomy, the introduction of edge AI computing enables drones to make real-time decisions without relying on external data processing. This allows for rapid adaptation to environmental changes, obstacle avoidance, and mission reconfiguration. Swarm intelligence is another transformative trend, allowing fleets of drones to collaborate autonomously using collective AI, thereby enhancing coordination,